

Scenario Based Assessment

INTRODUCTION:

Digital image enhancement improves image quality by reducing noise and preserving useful information. This report discusses two real-world scenarios and suitable enhancement techniques.

Scenario 1: Manufacturing Quality Inspection

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

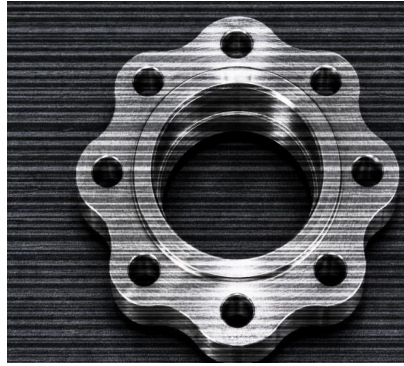
a) Interpret the scenario and identify the type of noise present.

The automated quality inspection system captures images of metal components. Due to electrical interference, repetitive horizontal stripes appear across the images. This type of degradation is called **Periodic Noise**.



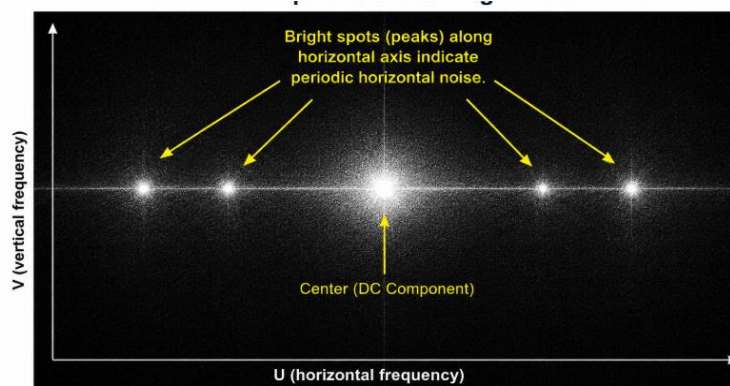
(b) Explain how this degradation affects inspection accuracy.

The horizontal stripes cover the surface details of the metal component, making it difficult to detect defects such as scratches, cracks, dents, holes, or surface irregularities.

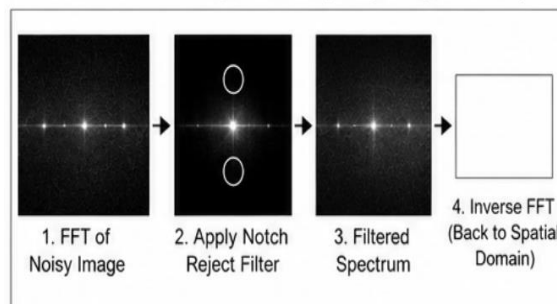


(c) Recommend an appropriate enhancement technique to remove the interference.

The most suitable enhancement technique is **Frequency Domain Filtering** using a **Notch Reject Filter**.



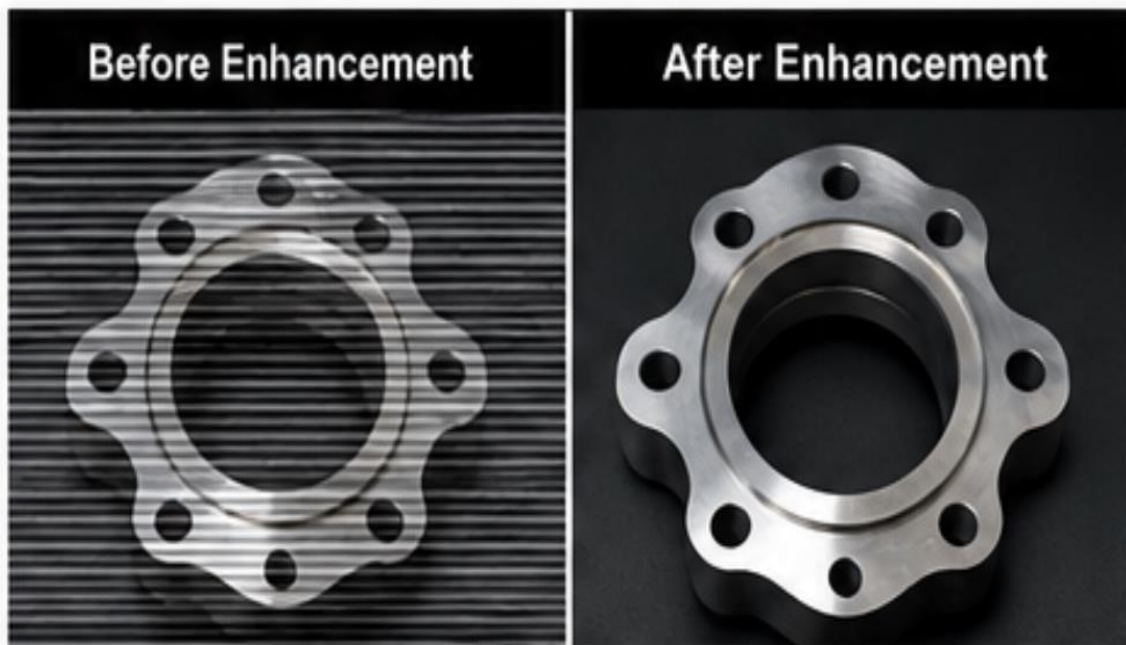
(d) A notch filter removes only the periodic frequency components while preserving the remaining image details. Spatial filters blur the entire image and are less effective for periodic noise.



Workflow of removing horizontal periodic noise using Notch Reject Filter in frequency domain.

(e) Logical Sequence:

1. Capture image.
2. Apply FFT.
3. Detect periodic peaks.
4. Apply Notch Reject Filter.
5. Perform Inverse FFT.
6. Inspect the enhanced image.



Comparison of noisy image and the enhanced image after applying Notch Reject Filter.

Scenario 2: Autonomous Vehicle Navigation

- (a) Heavy rain introduces noise while lane markings must remain sharp.

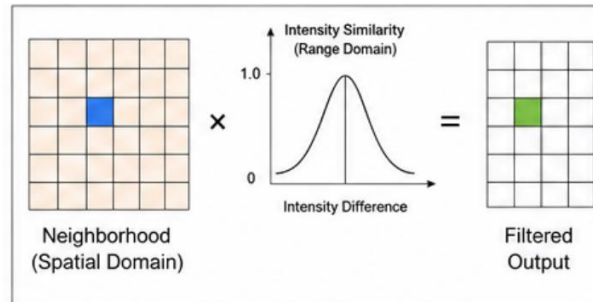


- (b) Preserving edges is important for lane detection, obstacle detection and safe navigation.



Lane markings and road edges detected. Preserving these edges is crucial for safe navigation.

- (c) Recommended technique: Bilateral Filter.



Bilateral filter combines spatial closeness and intensity similarity to preserve edges while removing noise.

(d) The bilateral filter smooths noise while preserving edges by considering pixel distance and intensity differences.

(e) Logical Sequence:

1. Capture road image.
2. Apply Bilateral Filter.
3. Preserve lane edges.
4. Detect lanes.
5. Safe navigation.



Bilateral filter removes noise while preserving important details like lane markings, road edges and objects.